



SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

MBA.Tech

Academic Year: 2022-2023

Program: B.Tech Stream ;Data Science

Year: II Semester: IV

Subject: Database Management Systems

Time: 3 hrs (10:00 am to 1:00 pm)

Date: 30 / 06 / 23

No. of Pages: 1 to 4

Marks: 100

Re- Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	[20]
CO-1 ; SO-1 ; BL-1	a.	Describe the three-schema architecture. Write down the difference between logical data independence and physical data independence?	[5]
CO-3 ; SO-1 ; BL-2	b.	Write down the difference between 3NF and BCNF. Explain with examples.	[5]
CO-1 ; SO-1 ; BL-1	c.	Write down the design issues of the Entity relationship model.	[5]
CO-4 ; SO-2 ; BL-2	d.	Explain ACID properties of database transactions.	[5]
Q2 CO-1; SO-1;	a.	Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and	[10]

BL-1		examinations conducted. List down all assumptions that you make about the mapping constraints.	
CO-1; SO-1; BL-2	b.	Write down the advantages and disadvantages of DBMS. List down the difference between a traditional file system and DBMS.	[10]
Q3 CO-2; SO-2; BL-3	a.	Consider the following relations for school database: STUDENT(sid, last name, firstname, major, entrydate, gpa, total credits, probation date) COURSE(course number, dept id, course name) DEPARTMENT(dept id, department name, address) INSTRUCTOR(instructor id, firstname,lastname, dept id) SECTION (section id, section number, course number, dept id, semester, instructor id) TRANSCRIPT(sid, section id, credits, grade) REQUIREMENT(req id, major, course number, dept id) Write following queries in SQL. i. Find the id and gpa of all students named "Eshanvi" (firstname). ii. Find the id of all courses offered by the department "Data Science". iii. Find the id of all sections of courses offered by the "Data Science" department in the "Fall99" semester. iv. Find the id of all sections that a student named "Eshanvi" is taking in "Fall99" semester. v. Find the last name and first name of all students who got an "A" at least once. vi. Find the identifier of all students who never got an "F". vii. Set the gpa of the student named "Eshanvi" to 5.0.	[14]

CO2-; SO-2; BL-3	b.	Distinguish between primary and super keys. Explain Referential Integrity constraint with suitable examples.	[6]
Q4 CO-3; SO-2; BL-3	a.	Define Normalization and Functional Dependency. Explain the first, second and third normal forms with examples.	[10]
CO-3; SO-1; BL-3	b.	Consider a relation R(A,B,C,D,E) with FDs {AB → C, DE → C, and B → D} i. Find the candidate keys of the above relation. ii. Indicate all BCNF violations for R. iii. Decompose the relations into collections of relations that are in BCNF. iv. Indicate which dependencies if any are not preserved by the BCNF decomposition.	[10]
Q5 CO-1; SO-1; BL-2	a.	Explain different types of attributes used in ER modeling with examples. Write down the features of Extended Entity-Relationship (EE-R) Model.	[10]
CO2-; SO-2; BL-3	b.	What is the difference between DDL, DML and DCL in SQL. Explain the following commands with examples in SQL. i. Alter ii. Select iii. Update iv. Drop v. Insert	[10]
Q6 CO-2; SO-1; BL-2	a.	Explain the following operators of relational algebra with example: i. Project, ii. Select, iii. Set difference	[10]

		iv Union v. Intersection	
CO-2; SO-1; BL-2	b.	Illustrate the different constraints available in SQL with examples of each.	[10]
Q7 CO-4; SO-1; BL-2	a.	Write down the advantages of NoSQL over traditional RDBMS and vice-versa?	[10]
CO-4; SO-1; BL-2	b.	Define transaction and discuss the states of transaction with suitable diagrams	[10]